

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Chip with 'Insane' AI inference performance

The AI hardware market is shifting with the introduction of Taalas' Hardcore AI architecture. The company has built a platform that converts AI



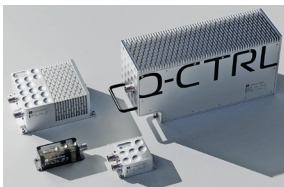
models into custom silicon by embedding the model, parameters, and

weights directly into hardware. This approach aims to deliver faster performance at lower cost, with new chips developed in about two months after a model is released. The company has designed hardware for a single AI model. Each chip is built specifically for one model, aiming to maximise performance and reduce cost. Early results suggest strong gains, but adoption will depend on whether large data centres are willing to move to such highly specialised systems.

Taalas
<https://taalas.com>

Quantum navigation system for GPS-free positioning

Q-CTRL's quantum navigation system, Ironstone Opal, is already tested across air, land, and maritime environments and is designed to provide positioning without relying on GPS. Ironstone



Opal uses quantum sensors and magnetic map matching to deliver

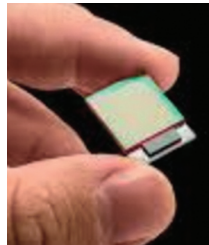
stable positioning over time, without drift. It works without external signals, making it resistant to jamming and spoofing, and usable in all visibility

conditions. The system also meets key aviation performance requirements, positioning it as a backup to GPS. The system is built to fit different platforms, including smaller UAVs, and can be deployed without restrictions across regions. The system operates alongside inertial navigation systems, significantly improving their accuracy while removing drift.

Q-CTRL
<https://q-ctrl.com>

Solid-state beam steering chip

Lumotive has announced the demonstration of a programmable two-dimensional (2D) photonic beamforming chip based on its light control meta-surface (LCM) architecture. This advance in semiconductor-based photonics enables electronic control of light across two dimensions within a single



chip, a capability pursued in optical science and considered essential for programmable optical systems. By enabling software-defined control of light, the technology opens opportunities in optical circuit switching for AI data centres, photonic communications, optical computing, and 3D sensing used in applications such as robotics. The chip forms and steers beams of light across two axes without moving parts.

Lumotive
<https://lumotive.com>

High-speed chip for complex AI workloads

Alibaba Group has introduced its CPU, XuanTie C950, for AI workloads. The company says the chip has been verified on a 5nm process. It did not name

the manufacturing partner, though sources indicate TSMC may have produced it. The chip is based on RISC-V, an open source architecture. The Xuan-



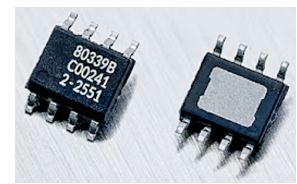
Tie C950 targets comparison with server-grade processors

as the company moves into the high-end segment. While RISC-V has grown in IoT and embedded systems, it has not yet entered high-performance computing at scale. The CPUs can be adjusted for specific inference needs, enabling users to tune them for their workloads. The chip can deliver more than 30% higher performance in targeted cases.

Alibaba Group
<https://www.alibaba.com>

Three-phase motor control chip

The MLX80339 offers three-phase motor control. It is part of Melexis' code-free IC range, with motor control logic built into the chip, allowing engi-



neers to configure systems without skills. Its code-free setup and

Start-to-Run tool reduce prototyping time. Designs can run in under 15 minutes using the evaluation kit and Melexis Compact Master. The tool generates configuration parameters for startup and operation, reducing engineering effort. Since the motor control logic is already tested by Melexis, engineers only need to complete system-level validation. It supports PWM-based fan control for drop-in replacement and also includes UART and I²C for diagnostics and multi-fan